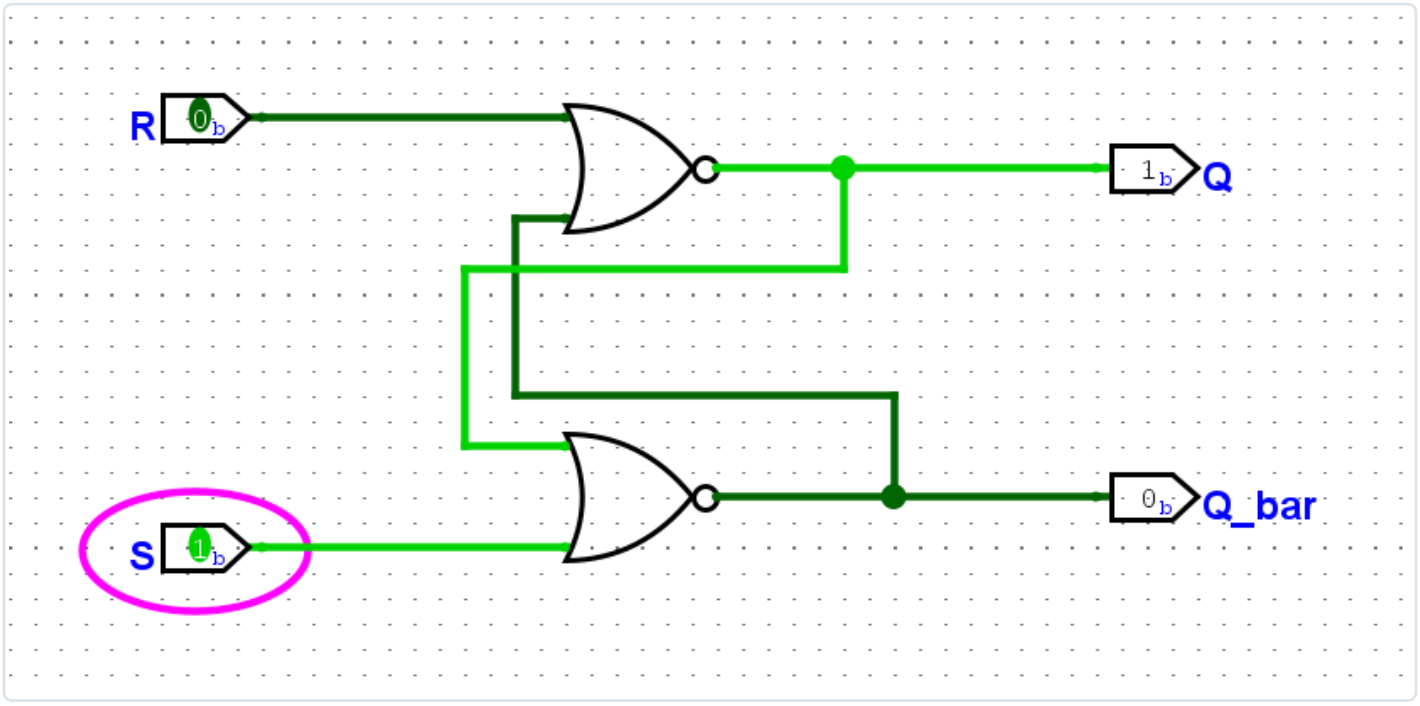
1. RS Latch (NOR)

Logisim gate-level circuit (student screenshot)



Brief explanation

Two cross-coupled NOR gates store one bit.
S=1 sets Q=1, R=1 resets Q=0, and S=R=0 retains the previous state.

Expected operating modes

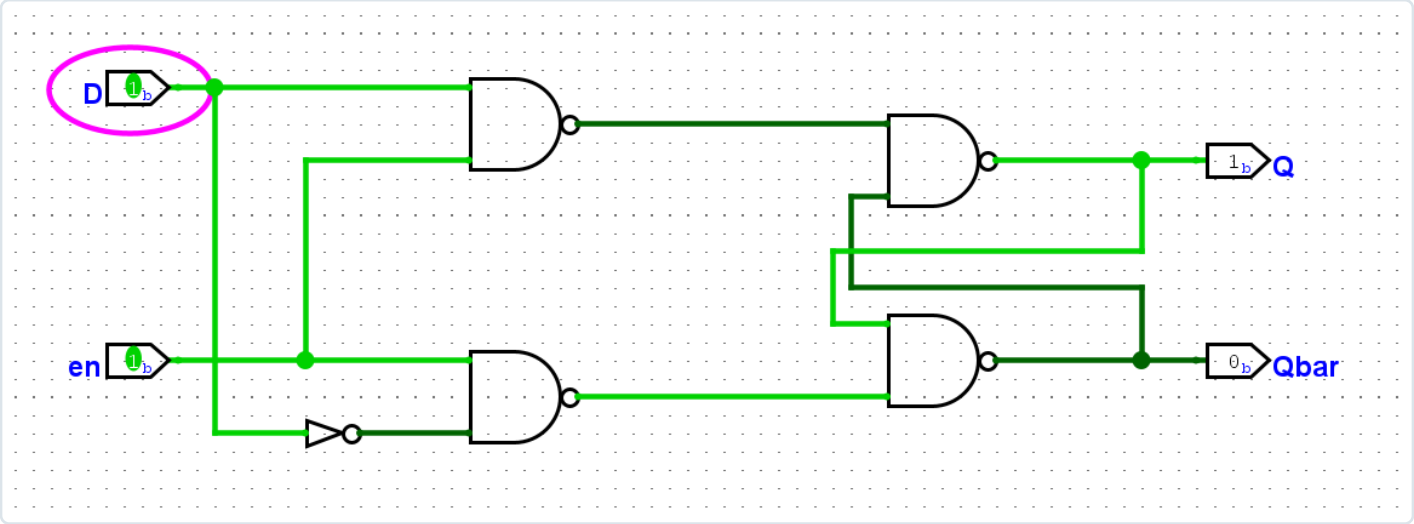
S	R	Q	Function
0	0	Previous Q	Hold
1	0	1	Set
0	1	0	Reset
1	1	0 (Qbar=0)	Forbidden

Corresponding VHDL design: rs_latch.vhd

The simultaneous release of the forbidden input is not guaranteed to recover to a unique state.

2. D Latch (NAND)

Logisim gate-level circuit (student screenshot)



Brief explanation

Four NAND gates and one inverter form a level-sensitive D latch.
When en=1, Q follows D. When en=0, Q keeps its previous value.

Expected operating modes

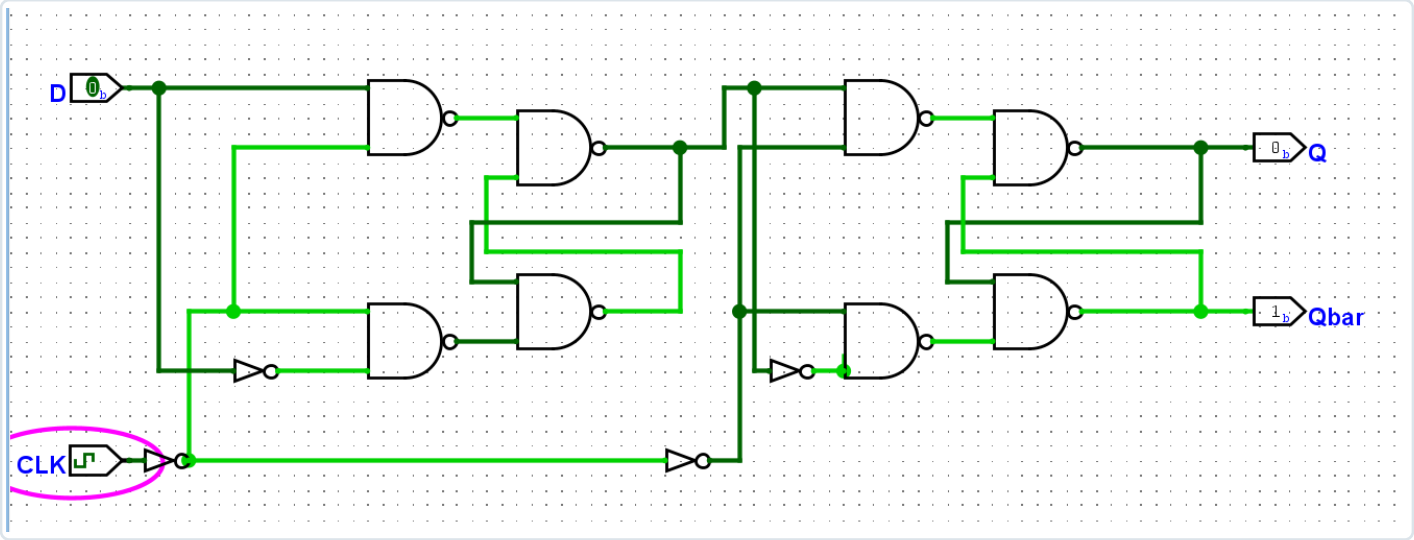
en	D	Q	Function
0	X	Previous Q	Hold
1	0	0	Transparent
1	1	1	Transparent

Corresponding VHDL design: d_latch.vhd

The source image is a sample with en=1 and D=1.

3. Positive-edge D Flip-Flop

Logisim gate-level circuit (student screenshot)



Brief explanation

Two gated D latches operate with opposite clock phases.
The master samples while CLK=0 and the slave transfers data when CLK rises.

Expected operating modes

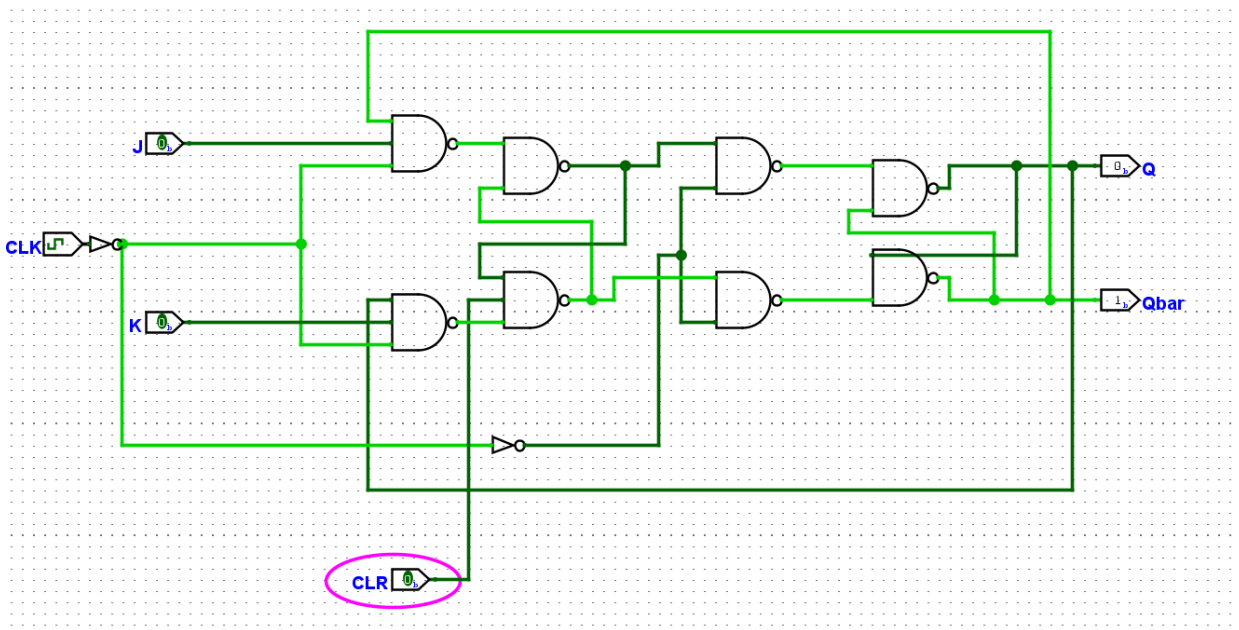
CLK condition	D	Q after event	Function
Rising edge	0	0	Load
Rising edge	1	1	Load
No rising edge	X	Previous Q	Hold

Corresponding VHDL design: d_flip_flop.vhd

The VHDL uses rising_edge(CLK), as required by the assignment.

4. JK Flip-Flop

Logisim gate-level circuit (student screenshot)



Brief explanation

NAND gates and Q/Qbar feedback implement JK state transitions.
CLR=0 initializes Q=0. During normal operation, CLR=1.

Expected operating modes

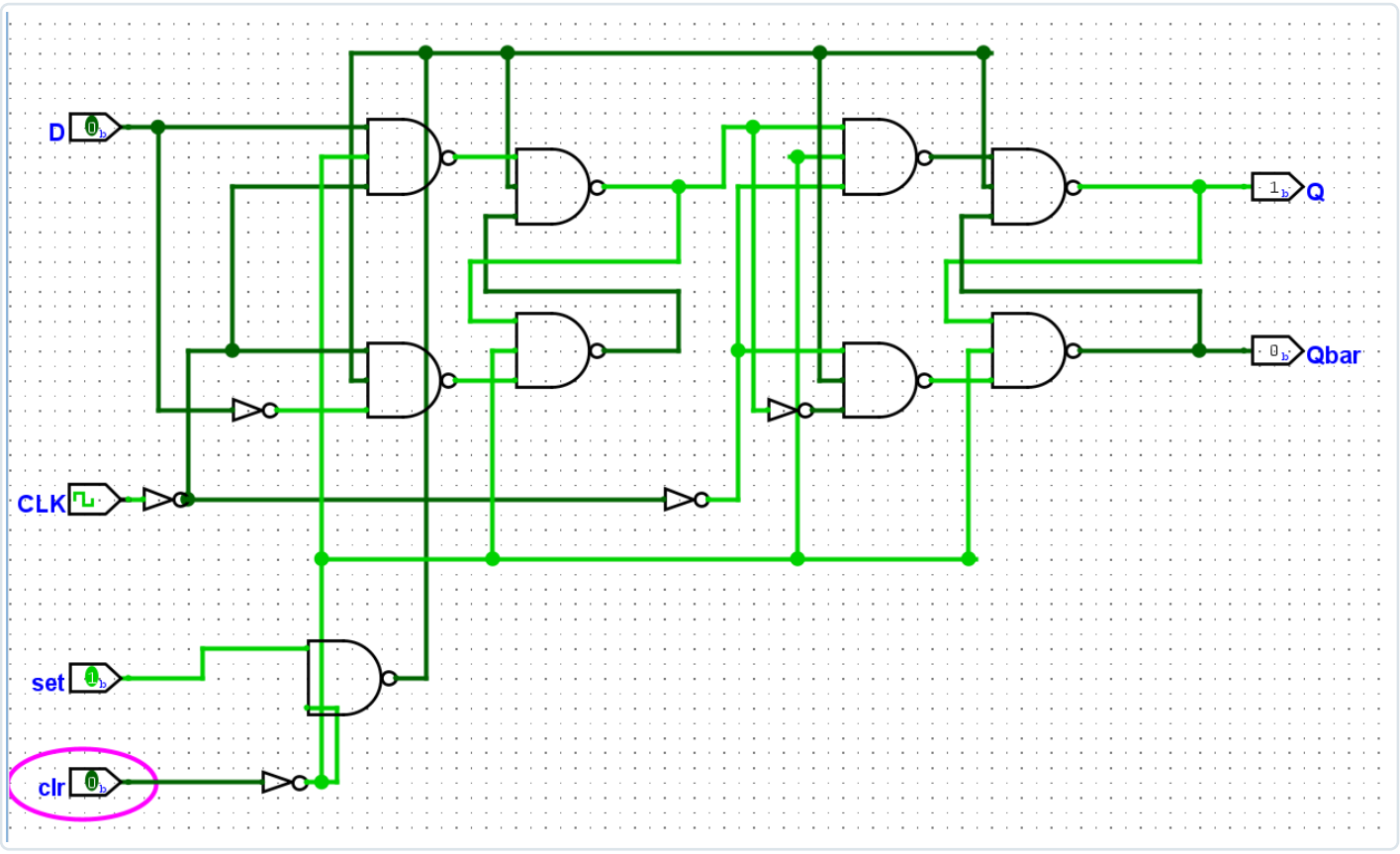
J	K	Next Q	Function
0	0	Previous Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	NOT Q	Toggle

Corresponding VHDL design: jk_flip_flop.vhd

Updated Logisim circuit screenshot supplied by the student. The capture shows CLR=0 and Q=0.
Set, Reset, Hold and Toggle must also be verified with CLR released (CLR=1).

5. D FF with Asynchronous Set/Clear

Logisim gate-level circuit (student screenshot)



Brief explanation

A rising-edge D flip-flop is extended with asynchronous control signals.
clr has higher priority than set, and both controls work independently of CLK.

Expected operating modes

clr	set	CLK	Q
1	X	X	0 immediately
0	1	X	1 immediately
0	0	Rising edge	D
0	0	Other	Previous Q

Corresponding VHDL design: d_flip_flop_async_set_clear.vhd

The image shows clr=1, set=0, and Q=0. Test set, both asserted, and normal sampling separately.